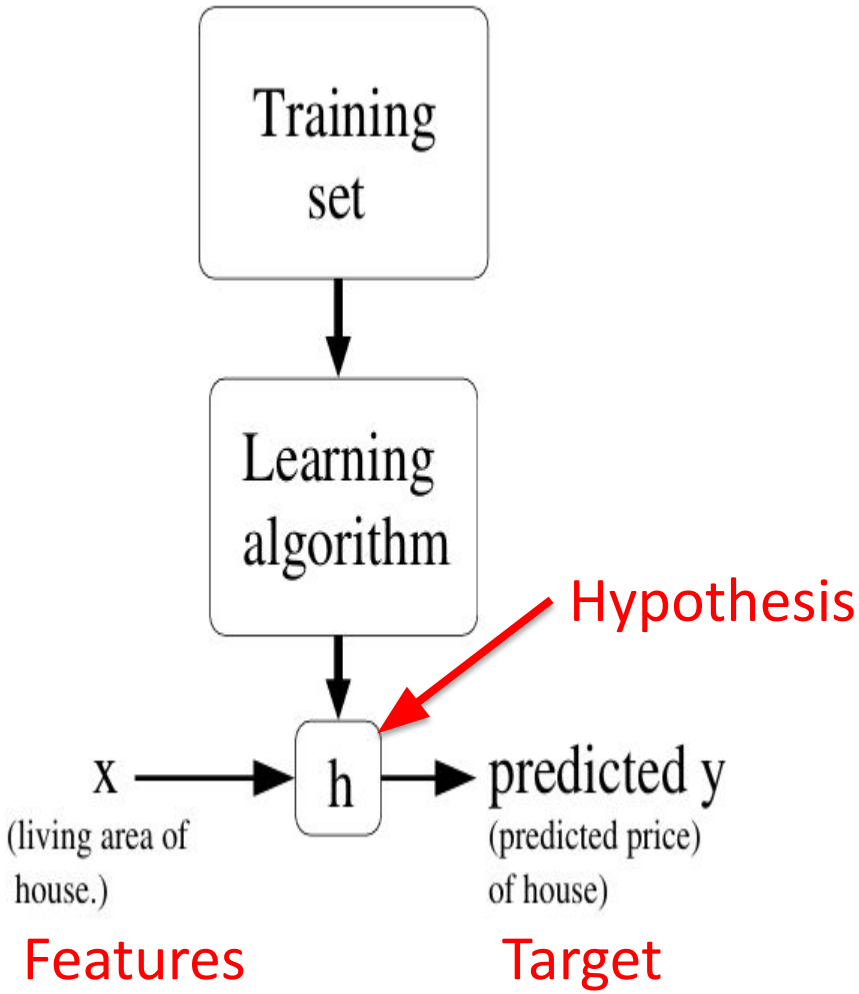
Linear Regression

CSE 329

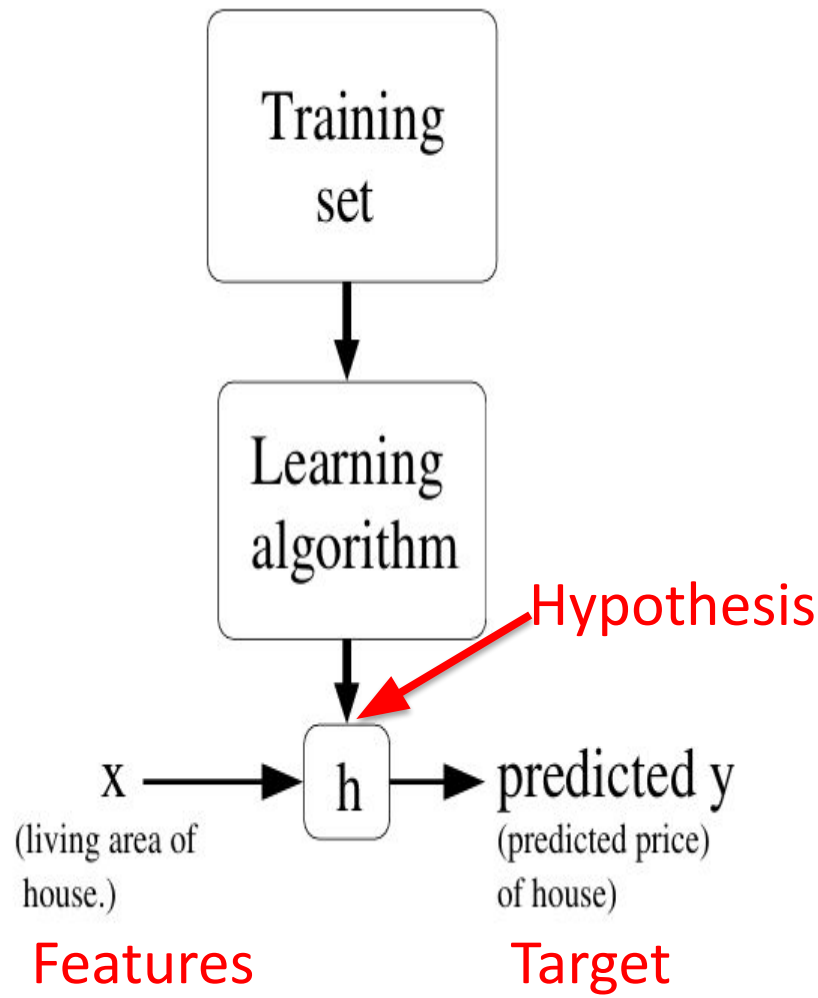
Supervised Learning (Recap)



Supervised Learning

- $x^{(i)}$: input or feature
- $y^{(i)}$: output or target
- $(x^{(i)}, y^{(i)})$: i^{th}training example
- $\{(x^{(i)}, y^{(i)}): i = 1,2, \dots, n\}$ the training set

- Goal??



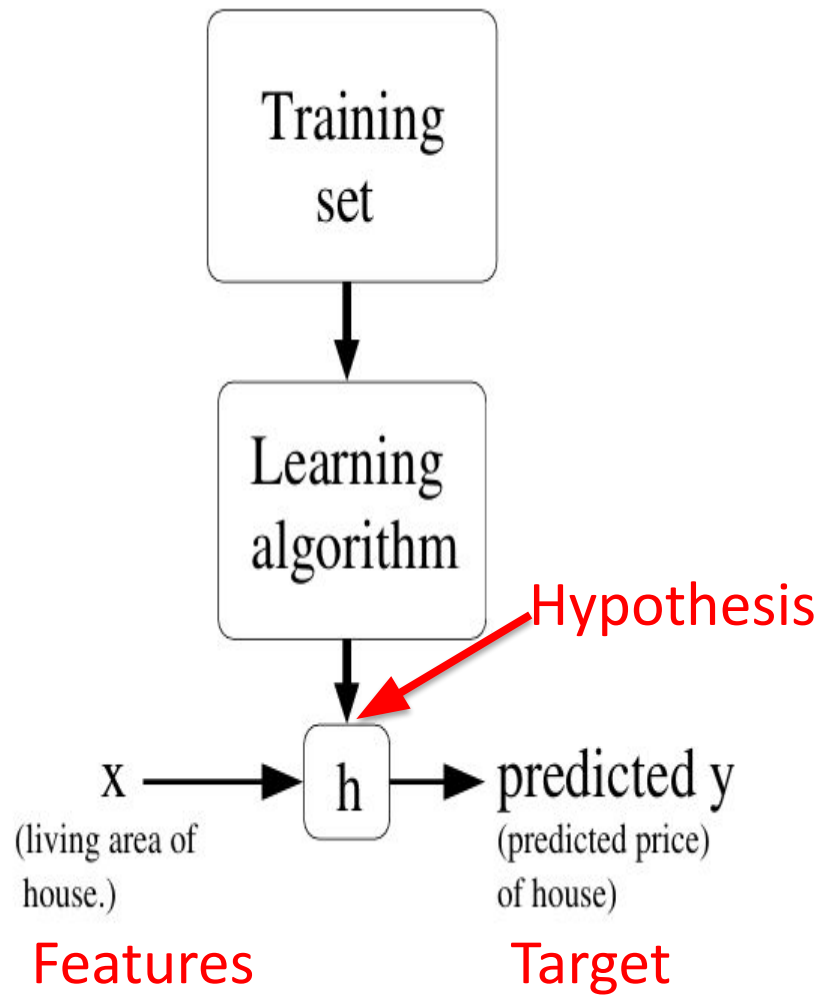
Supervised Learning

- **X** is the input space
- **y** is the target space
- Learn a function or hypothesis such that

$$h: X \rightarrow Y$$

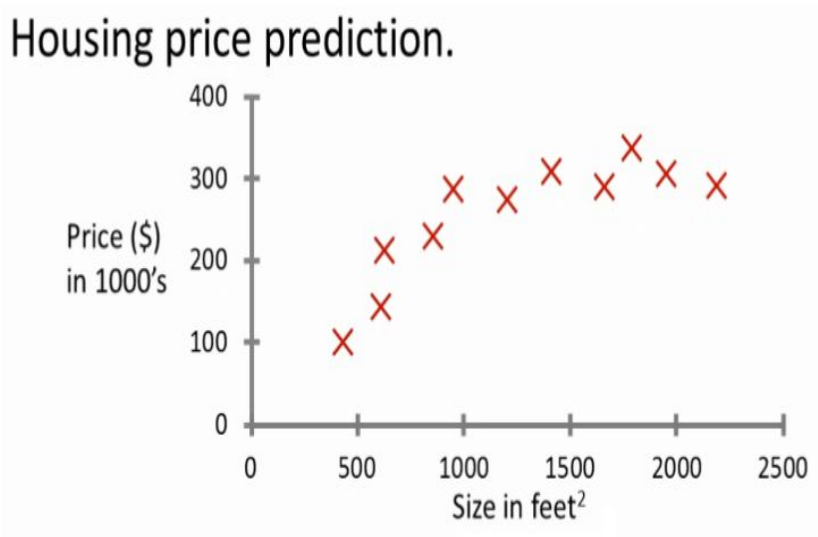
i.e.

$$h(x) = y$$



Housing Price Prediction

Regression !!!



Living area (feet^2)	Price (1000$s)
2104	400
1600	330
2400	369
1416	232
3000	540
⋮	⋮

Learn a function h such that

h(living area in $feet^2$) ~ Price in 1000$s

Regression vs Classification

- Regression
  - Target variable is continuous
  - Example,
    - Housing price prediction
- Classification
  - Target variable can take only small number of discrete values
  - Example,
    - Email classification (Spam vs Ham)

Linear Regression

- We assume a linear relation between X and Y.
- Given x, we approximate y with a linear function.

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

$x_0 =$

$$h_\theta(x) = \sum_{i=0}^{d} \theta_i x_i = \theta^T x$$

- θ is called parameters/weights
  - A d+1 dimensional vector of weights

- Housing price prediction with two features

Living area (feet^2)	#bedrooms	Price (1000$s)
2104	3	400
1600	3	330
2400	3	369
1416	2	232
3000	4	540
⋮	⋮	⋮

Linear Regression

- Goal:
  - Given a training set, learn θ
  - Such that $h_\theta(x)$ is as close as possible to y

- If we had only one sample, minimize

$$J(\theta) = \frac{1}{2}\left(h_\theta(x) - y\right)^2$$

$$h_\theta(x) = \sum_{i=0}^{d} \theta_i x_i = \theta^T x$$

Linear Regression

- Goal:
  - Given a training set, learn θ
  - Such that $h_\theta(x)$ is as close as possible to y

Least Squares (LS)

- For a dataset with n samples, minimize

$$J(\theta) = \frac{1}{2}\sum_{k=1}^{n}\left(h_\theta(x^{(k)}) - y^{(k)}\right)^2$$

Cost Function/Loss Function

$$h_\theta(x) = \sum_{i=0}^{d}\theta_i x_i = \theta^T x$$

Linear Regression

- Goal: Choose θ to minimize $\mathrm{J}(\theta)$

$$\theta^* = argmin_\theta J(\theta)$$

- How?

Linear Regression

$$J(\theta) = \frac{1}{2}\sum_{k=1}^{n}\left(h_\theta(x^{(k)}) - y^{(k)}\right)^2$$

- Goal: Choose θ to minimize $J(\theta)$

$$\theta^* = argmin_\theta J(\theta)$$

- Set partial derivatives to 0 to obtain minima

$$\frac{\partial J}{\partial \theta_0} = \sum_{k=1}^{n}(\theta_0 + \theta_1 x_1^{(k)} + \theta_2 x_2^{(k)} - y^{(k)}) = 0$$

$$\frac{\partial J}{\partial \theta_1} = \sum_{k=1}^{n}(\theta_0 + \theta_1 x_1^{(k)} + \theta_2 x_2^{(k)} - y^{(k)})x_1^{(k)} = 0$$

$$\frac{\partial J}{\partial \theta_2} = \sum_{k=1}^{n}(\theta_0 + \theta_1 x_1^{(k)} + \theta_2 x_2^{(k)} - y^{(k)})x_2^{(k)} = 0$$

Linear Regression

- Simplifying the partial derivative equations,

$$\sum_{k=1}^{n} \left(\theta_0 + \theta_1 x_1^{(k)} + \theta_2 x_2^{(k)} - y^{(k)}\right) = 0$$

$$\Rightarrow \quad n\theta_0 + \theta_1 \sum_k x_1^{(k)} + \theta_2 \sum_k x_2^{(k)} = \sum_k y^{(k)}$$

Linear Regression

- Simplifying the partial derivative equations,

$$n\theta_0 + \theta_1 \sum_k x_1^{(k)} + \theta_2 \sum_k x_2^{(k)} = \sum_k y^{(k)}$$

$$\sum_{k=1}^{n} \left(\theta_0 + \theta_1 x_1^{(k)} + \theta_2 x_2^{(k)} - y^{(k)}\right) x_1^{(k)} = 0$$

$$\Rightarrow \quad \theta_0 \sum_k x_1^{(k)} + \theta_1 \sum_k (x_1^{(k)})^2 + \theta_2 \sum_k x_1^{(k)} x_2^{(k)} = \sum_k x_1^{(k)} y^{(k)}$$

Linear Regression

- Simplifying the partial derivative equations,

$$n\theta_0 + \theta_1 \sum_k x_1^{(k)} + \theta_2 \sum_k x_2^{(k)} = \sum_k y^{(k)}$$

$$\theta_0 \sum_k x_1^{(k)} + \theta_1 \sum_k (x_1^{(k)})^2 + \theta_2 \sum_k x_1^{(k)} x_2^{(k)} = \underline{\sum_k x_1^{(k)} y^{(k)}}$$

$$\sum_{k=1}^{n} \left(\theta_0 + \theta_1 x_1^{(k)} + \theta_2 x_2^{(k)} - y^{(k)}\right) x_2^{(k)} = 0$$

$$\Rightarrow \quad \theta_0 \sum_k x_2^{(k)} + \theta_1 \sum_k x_1^{(k)} x_2^{(k)} + \theta_2 \sum_k (x_2^{(k)})^2 = \underline{\sum_k x_2^{(k)} y^{(k)}}$$

Linear Regression

- Simplifying the partial derivative equations,

$$n\theta_0 + \theta_1 \sum_k x_1^{(k)} + \theta_2 \sum_k x_2^{(k)} = \sum_k y^{(k)}$$

$$\theta_0 \sum_k x_1^{(k)} + \theta_1 \sum_k (x_1^{(k)})^2 + \theta_2 \sum_k x_1^{(k)} x_2^{(k)} = \sum_k x_1^{(k)} y^{(k)}$$

$$\theta_0 \sum_k x_2^{(k)} + \theta_1 \sum_k x_1^{(k)} x_2^{(k)} + \theta_2 \sum_k (x_2^{(k)})^2 = \sum_k x_2^{(k)} y^{(k)}$$

System of Linear Equations

Linear Regression

- Simplifying the partial derivative equations,

$$n\theta_0 + \theta_1 \sum_k x_1^{(k)} + \theta_2 \sum_k x_2^{(k)} = \sum_k y^{(k)}$$

$$\theta_0 \sum_k x_1^{(k)} + \theta_1 \sum_k (x_1^{(k)})^2 + \theta_2 \sum_k x_1^{(k)} x_2^{(k)} = \sum_k x_1^{(k)} y^{(k)}$$

$$\theta_0 \sum_k x_2^{(k)} + \theta_1 \sum_k x_1^{(k)} x_2^{(k)} + \theta_2 \sum_k (x_2^{(k)})^2 = \sum_k x_2^{(k)} y^{(k)}$$

- $$\begin{pmatrix} n & \sum x_1^{(k)} & \sum x_2^{(k)} \\ \sum x_1^{(k)} & \sum (x_1^{(k)})^2 & \sum x_1^{(k)} x_2^{(k)} \\ \sum x_2^{(k)} & \sum x_1^{(k)} x_2^{(k)} & \sum (x_2^{(k)})^2 \end{pmatrix} \begin{pmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} \sum y^{(k)} \\ \sum x_1^{(k)} y^{(k)} \\ \sum x_2^{(k)} y^{(k)} \end{pmatrix}$$

Linear Regression

- Simplifying the partial derivative equations,

$$n\theta_0 + \theta_1 \sum_k x_1^{(k)} + \theta_2 \sum_k x_2^{(k)} = \sum_k y^{(k)}$$

$$\theta_0 \sum_k x_1^{(k)} + \theta_1 \sum_k (x_1^{(k)})^2 + \theta_2 \sum_k x_1^{(k)} x_2^{(k)} = \sum_k x_1^{(k)} y^{(k)}$$

$$\theta_0 \sum_k x_2^{(k)} + \theta_1 \sum_k x_1^{(k)} x_2^{(k)} + \theta_2 \sum_k (x_2^{(k)})^2 = \sum_k x_2^{(k)} y^{(k)}$$

- Vectorized Form

$$\begin{pmatrix} n & \sum x_1^{(k)} & \sum x_2^{(k)} \\ \sum x_1^{(k)} & \sum (x_1^{(k)})^2 & \sum x_1^{(k)} x_2^{(k)} \\ \sum x_2^{(k)} & \sum x_1^{(k)} x_2^{(k)} & \sum (x_2^{(k)})^2 \end{pmatrix} \begin{pmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} \sum y^{(k)} \\ \sum x_1^{(k)} y^{(k)} \\ \sum x_2^{(k)} y^{(k)} \end{pmatrix}$$

$$(X^T X)\theta = X^T y$$

X is a $n \times (d+1)$ matrix
Each row is a multivariate sample

y is a $n \times 1$ vector
Each row is a label of a sample

Linear Regression

- Solve the system of linear equations to get θ^*

$$(X^T X)\theta = X^T y$$

$$\theta^* = (X^\top X)^{-1} X^\top y$$

Normal Equation

Linear Regression

- Once trained predict housing prices with,

$$\hat{y} = (\theta^*)^T x$$

- What if n >> 1 ??
- What if d >> 1 ??
- What if the $(X^T X)$ is non-invertible?
- What if the relation between **X** and **y** are not linear?

Reference

- Dive into Deep Learning (d2l)
  - Sections 3.1 (up to 3.1.2), 3.4

- Deep Learning [Optional Reading]
  - Section 5.1.4